
UNIVERSITY OF BRISTOL

Summer 2025 Examination Period

School of Engineering Mathematics and Technology

Coursework for the Degree of Master of Engineering

UNIT CODE SEMTM0009

Mathematical Modelling in Biology, Medicine and Public Health

Deadline: 11 March, 2026

Instructions

- This coursework comprises 4 questions.
- Each question carries 25 marks.
- Answer all questions.
- Your answers should be submitted as a report no longer than 16 pages (including references). Note that a well-written document with significantly fewer than this is also acceptable —this is a strict upper bound, not a target! Any pages above the 16th one will not be assessed. The final report should be submitted as a pdf file per the template on BlackBoard.
- You should also explain the method you used to perform the numerical parameter fits, upload your code to a GitHub repo that you should share with `edgardeitor`.

Use of AI

You may only use tools such as spelling and grammar checkers in this assignment, and their use should be limited to corrections of your own work rather than substantial rewrites or extended contributions.

Regarding use of AI technology, please refer to Bristol guidelines:

<https://www.bristol.ac.uk/students/support/academic-advice/using-artificial-intelligence/>

Uses of tools outside of these limitations may incur academic misconduct penalties. You are not obligated to use AI tools in your work, and no additional credit is awarded for their use.

Q1. The SEIR and SIR models are two widely used frameworks for modelling infectious disease dynamics. They are given by

SEIR model

$$\begin{aligned}\frac{dS}{dt} &= dN - \beta_1 \frac{SI}{N} - dS, \\ \frac{dE}{dt} &= \beta_1 \frac{SI}{N} - aE - dE, \\ \frac{dI}{dt} &= aE - g_1 I - dI, \\ \frac{dR}{dt} &= g_1 I - dR,\end{aligned}$$

SIR model

$$\begin{aligned}\frac{dS}{dt} &= dN - \beta_2 \frac{SI}{N} - dS, \\ \frac{dI}{dt} &= \beta_2 \frac{SI}{N} - g_2 I - dI, \\ \frac{dR}{dt} &= g_2 I - dR,\end{aligned}$$

where the E in SEIR stands for the Exposed population, who are infected but still in a latent phase (they cannot infect others).

In both models, the total population size remains constant at a level N . The average time from infection to recovery is fitted to epidemiological data and is assumed to be the same in both models.

- a) For each model, calculate the basic reproduction number, R_0 . (3 marks)
- b) For each model, find the non-trivial (endemic) equilibrium, and compute the Jacobian matrix evaluated at that equilibrium. (5 marks)
- c) For each model, determine the early growth rate of infection following the invasion of a small number of infected individuals. (5 marks)
- d) Show that, if the two models are parameterised such that they have the same basic reproduction number R_0 , then:
 - (i) the equilibrium level of susceptible individuals is the same in both models; (2 marks)
 - (ii) the number of new infections per unit time at equilibrium is the same in both models; (3 marks)
 - (iii) the early exponential growth rate of infection is slower in the SEIR model (you may assume that d is negligible compared to all other rates.) (7 marks)

Q2. Smallpox can be considered a disease with SEIR-type dynamics and a level of mortality, m , following infection. Assuming that the birth and death rates are such that the population size is constant leads to the following differential equations:

$$\begin{aligned}\dot{S} &= d - \beta SI - dS, \\ \dot{E} &= \beta SI - aE - dE, \\ \dot{I} &= aE - gI - dI, \\ \dot{R} &= g(1 - m)I - dR.\end{aligned}$$

(Note that we assume density-dependent transmission to avoid complications with changing population size with mortality).

- a) Calculate the basic reproduction number, R_0 , for this model, and explain why it is unaffected by the mortality probability, m . (3 marks)
- b) A decision is made to vaccinate a proportion, p , of newborn individuals against smallpox, but there is a probability, q , of dying due to the vaccine. Write down the new model and find the critical level of vaccination. Assume that vaccination is perfectly effective and immediate. (5 marks)
- c) Using the model from part (b), find the steady state solution for the number of infected individuals when a proportion p is vaccinated at birth and thus show that the number of deaths due to vaccination is lower than the number of deaths due to infection when

$$q < mg \frac{[(1 - p)R_0 - 1]}{p\beta}.$$

(10 marks)

- d) Given that vaccination is maintained at a level that keeps smallpox eradicated (no deaths from smallpox), find an expression for the minimum number of deaths over a given period of time, Δt , in terms of R_0 . Comment on the rationale for recommending the smallpox vaccine as a public health policy even if $q > 0$. (7 marks)

Q3. A disease is described by the SIR model:

$$\begin{aligned}\frac{dS}{dt} &= d - \beta SI - dS, \\ \frac{dI}{dt} &= \beta SI - gI - dI, \\ \frac{dR}{dt} &= gI - dR,\end{aligned}$$

where S , I , and R refer to proportions of the population.

- a) Formulate a new model that includes vaccination of a proportion, p , of all newborn individuals. (3 marks)
- b) Show that the equilibrium level of infection and the growth rate of infection in a population at the disease-free equilibrium both reach zero at the same critical level of vaccination. (3 marks)

Both vaccination and infection can cause medical problems, such that individuals need to be hospitalised with probabilities h_V and h_I , respectively (you may assume that $h_V < h_I$.)

- c) What is the optimum level of vaccination, p^* , that minimises the hospitalisation rate in the entire population? (4 marks)
- d) Given that the rest of the population is vaccinating at level p , find a probability of vaccination, q , that minimises an individual's risk of hospitalisation. (4 marks)
- e) A given vaccination level p is an evolutionary stable strategy if the risk of hospitalisation is the same for all q . Find this evolutionary stable strategy, and show that it is less than the optimum for the entire population. Interpret the meaning of the different levels of vaccination in parts c) – e) of this question. (7 marks)
- (f) Consider an imperfect vaccine given at birth. Explain how your model would change to account for these two scenarios, separately:
 - i. A proportion π of the vaccinated population is perfectly protected. The remaining vaccinated population is not protected. (2 marks)
 - ii. Each vaccinated individual has a probability π of being protected by the vaccine. (2 marks)

Q4. Consider an SEIR model given by

$$\begin{aligned}\dot{S} &= -\beta S I, \\ \dot{E} &= \beta S I - \sigma E, \\ \dot{I} &= \sigma E - \gamma I, \\ \dot{R} &= \gamma I,\end{aligned}$$

where the parameters can be understood as follows:

β : transmission rate,
 σ : incubation rate,
 γ : recovery rate.

Furthermore, assume that $S(0) + E(0) + I(0) + R(0) = 1$.

In a real scenario, not all the variables are observable. In this case, we can only observe $y(t) = \rho \sigma E(t)$, which is a proportion of new infections, where $\rho \in (0, 1]$ is a reporting fraction. If we assume that we are at an early stage of the infection, i.e., $S(t) \approx S(0) \approx 1$, then,

- (i) Find the equation for $y(t)$ and solve it, assuming that $y(0) = 0$, and $y'(0) = 1$.
HINT: Find $\dot{y}$, and $\ddot{y}$ and relate them, together with y . (5 marks)
- (ii) What parameters are identifiable from y ? Justify your answer. (3 marks)
- (iii) An expert in epidemiology has data for a specific epidemic. They ask you to design and explain a method for fitting the identifiable parameters of your setting, and find approximations of these parameter values by using the data provided on Blackboard in the dataset `data.csv`. This dataset has two columns, the first one for time t , measured in days, and the second one for a noisy dataset obtained from $(y(t))^2$. Comment on the epidemiological significance of the parameter values you obtained. (8 marks)
- (iv) As this is only an approximation for t being small, this expert also asks you to fit an exponential function $g(t) = a + b e^{ct}$ to the same dataset. Find also a suitable polynomial function to fit the data. Explain how these fittings compare to the one carried out for item (iii). Comment on the robustness of the parameter values you obtained in terms of using them to inform public health policy. (9 marks)